

Dawei Xiong

Ph.D. Student

The Hong Kong University of Science and Technology (Guangzhou)

mailto:dxiong107@connect.hkust-gz.edu.cn (Email)

Research Interests

Behavioral Public Administration | AI Agent | Virtual Reality | Public Service

About

I am currently a Ph.D. student at HKUST(GZ) Society Hub, Innovation, Policy and Entrepreneurship Thrust. My research interests include behavioral public administration, AI Agent, Virtual Reality (VR), and public service.

Education

HKUST(GZ) — Ph.D. in Innovation, Policy and Entrepreneurship

2025.9 – Present

Guangzhou, China

SUPERVISORS Xun Wu · Xu Zhang

Congratulations to Prof. Xun Wu and Prof. Mingming Fan on being named in Stanford's Top 2% Scientists List 2026!

HKUST(GZ) — M.Phil. in Innovation, Policy and Entrepreneurship

2023.9 – 2025.7

Guangzhou, China

SUPERVISORS Xun Wu · Mingming Fan

Congratulations to Prof. Xun Wu and Prof. Mingming Fan on being named in Stanford's Top 2% Scientists List 2026!

Sun Yat-sen University — B.S. in Applied Psychology

2012.9 – 2016.7

Guangzhou, China

SUPERVISOR Lehua Cheng

Latest book: 《感性理性系统分化说：情理关系的重构》

Work Experience

YUANHUO Technology — Co-founder & COO

2024.5 – Present

Guangzhou, China

- Company focus: multi-modal AI agent solutions for digital education and marketing.
- "LLM-driven digital friends and communities for you" won **2025 CES Innovation Award** (only one from mainland China, shortest time to exist).
- Responsibilities: business development, process optimization, internal operations management.

Related Report: Shining! HKUST(GZ) Faculty & Alumni Startups Win the CES Innovation Award 2025

HKUST BLUEBAY Incubator — Supervisor / Officer / Assistant

2016.12 – 2022.6

Shenzhen, China

- Organized HKUST One Million Dollar Competition (Shenzhen) for 5 years (2017–2021).
- Helped startups refine business plans and connect with industrial partners.
- Since 2020, built and operated BLUEBAY Incubator's Futian Base in the Hetao Shenzhen-Hong Kong Innovation Cooperation Zone; incubated 20 enterprises.

Research

PUBLICATIONS

Cognitive Load as a Hidden Cost of Empathy: Tradeoffs in Using VR for Inclusive Design

Empathy & Inclusive Design · Public Service · Human-Computer Interaction · VR Experiment

Xiong, D., Ma, Z., Ling, H., Wu, X., Fan, M., & Zhang, X.

International Journal of Human–Computer Interaction, 2026

Indexed in SCIE & SSCI · 2025 Journal Impact Factor 6.1 · JCR Q1 (Computer Science—Cybernetics, Ergonomics)

Virtual reality (VR) is used to bridge the "empathy gap" in inclusive design, but whether emotionally evocative simulations improve design outcomes remains unclear. We conducted a between-subjects experiment comparing passive video with non-emotional and emotional interactive VR simulations of low-vision mobility challenges, informed by interviews and co-design with low-vision participants. Emotional intensity was manipulated via non-player-character feedback while task structure and task-relevant information were held constant. We measured empathy change and assessed design ideas by alignment with user-prioritized needs. Both VR conditions increased empathy relative to video, but emotional VR produced no additional measurable gain, whereas non-emotional VR produced stronger user-need-aligned ideation than both other conditions. Drawing on cognitive load theory, we propose that emotionally charged cues may introduce extraneous demands during ideation, offering a working hypothesis for this divergence. Our work provides guidelines for calibrating VR emotional intensity to balance empathic engagement and design ideation.

Full Text (DOI) (https://doi.org/10.1080/10447318.2026.2730903) **Open Access (ScienceDB) (https://doi.org/10.57760/sciencedb.45738)**

Plain-Language Summary (https://mp.weixin.qq.com/s/4Z9jU7JK0dcvr9cvZhAqaQ)

PUBLICATION HISTORY

Earlier titled "Enhancing Empathy Through Virtual Reality: A Behavioral Experiment on Understanding Service Users with Visual Impairments".

- 🎤 **Presented, April 2025**
  International Research Society for Public Management (IRSPM) Conference, Bologna, Italy
- ❌ **Rejected, May 2025**
  Journal of Business Research (JBR)
- 🎤 **Presented, June 2025**
  Economic Science Association (ESA) World Meeting, Beijing, China
- 🏆 **First Prize for Excellent Paper, November 2025**
  2025 Lingnan Academic Forum
- ❌ **Rejected in Second Round, December 2025**
  CHI 2026 — ACM Conference on Human Factors in Computing Systems
- ✅ **Accepted, September 2026**
  International Journal of Human–Computer Interaction (IJHCI)

Investigating Size Congruency Between the Visual Perception of a VR Object and the Haptic Perception of Its Physical World Agent

Haptic-Visual Congruency · Visualization & Interaction · VR Interaction Study

Zheng, W., **Xiong, D.**, Li, J., Jiang, J., Weng, C., Zhou, J., & Fan, M.

17th International Symposium on Visual Information Communication and Interaction (VINCI), 2024, pp. 1–9

The perception of physical objects and miniatures enhances the realism and immersion in VR. This work explores the relationship between haptic feedback from real objects and their visual representations in VR. The study examines how users confirm and adjust the sizes of different virtual objects. The results show that as the size of the virtual cubes increases, users are less likely to perceive the size correctly and need more adjustments. This research provides insights into how haptic sensations and visual inputs interact, contributing to the understanding of visual-haptic illusions in VR environments.

Full Text (DOI) (https://doi.org/10.1145/3678698.3678706) **PDF (arXiv) (https://arxiv.org/pdf/2408.08018)**

ONGOING PROJECTS

Unbundling incubation service research

Entrepreneurship Incubation · *Organization Science · Embedded Multiple-case Study*

Examines how business incubation services are unbundled across organizational boundaries, and how specialized providers, platforms, and startups divide the work of integrating them.

Research in Legislative Debate

Legislator behavior research · *Politics · Grounded Theory*

Examines how legislators reason through and handle public-value trade-offs, using constructivist grounded theory on official records from three Hong Kong Legislative Council terms.

Recombination and Coordination Processes in Multidisciplinary Academic Entrepreneurial Teams: Explaining Innovative New Venture Idea Generation

Multi-discipline Co-operation · *Organization Science · LLM Agent-based Modeling*

Opens the process black box between academic entrepreneurial team composition and early-stage new venture idea generation. Using LLM-based agent simulation, it traces how disciplinary differences expand recombination potential while simultaneously creating coordination friction — representing team members with different disciplinary orientations and generating complete interaction traces of how ideas are proposed, challenged, revised and recombined.

AI Court — Who Is Responsible When AI Gets It Wrong?

AI Liability Adjudication · *Public Administration · Human-AI Panel Review*

A public adjudication platform where a three-reviewer panel (two human reviewers and one AI reviewer) rules on whether an AI deserves to be held responsible for what it did — turning real-world AI failures into an open, citable case law.

Try it now! ai-court.xion516.com (https://ai-court.xion516.com)

Teaching

Principles of Economics

Fall 2026

UFUG 1801 Teaching Assistant

Research Design for Innovation, Policy and Entrepreneurship Studies

Spring 2026

IPEN5120 Teaching Assistant

Introduction to Technological Innovation and Social Entrepreneurship

Fall 2025

SOCH 5000 Teaching Assistant

Cross-disciplinary Design Thinking

Spring 2025

UCMP 6030 Teaching Assistant

All courses taught at The Hong Kong University of Science and Technology (Guangzhou).